UNIT 2: SHAPES AND REGIONS

SESSION 6: SHAPE MODELS AND SHAPE RECOGNITION

SLO 2: APPLY SHAPE RECOGNITION TECHNIQUES TO CLASSIFY OR IDENTIFY OBJECTS BASED ON THEIR GEOMETRIC FEATURES.

Shape Feature Identification

Observe the following geometric features of objects and identify the most likely shape.

Area	Aspect Ratio	Boundary Shape	Identified Shape
Medium	1 : 1	Smooth curve	Circle
Large	2 : 1	Straight edges	Rectangle
Small	1 : 3	Long and narrow	Rectangle

Shape Classification

Three objects are detected in a binary image.

Object	Area (pixels)	Aspect Ratio	Shape Hint	Classification
A	50	1 : 1	Rounded boundary	Circle
B	80	3 : 1	Long rectangular shape	Rectangle
C	40	1 : 1	Straight edges	Square

Activity 3 – Feature Matching

Match the shape feature with its purpose**.**

Shape Feature	Purpose
Area	A. Measures object boundary length
Perimeter	B. Determines object size
Aspect Ratio	C. Identifies elongation of shape
Compactness	D. Measures how circular an object is

Write the correct matches:

Area → B

Perimeter → A

Aspect Ratio → C

Compactness → D